

torch.nn.functional.binary_cross_entropy

https://pytorch.org/docs/stable/generated/torch.nn.functional.binary_cross_entropy

Description:

Compute Binary Cross Entropy between the target and input probabilities. See BCELoss for details. input (Tensor) – Tensor of arbitrary shape as probabilities. target (Tensor) – Tensor of the same shape as input with values between 0 and 1. weight (Tensor, optional) – a manual rescaling weight if provided it's repeated to match input tensor shape

Function Signature

```
torch.nn.functional.binary_cross_entropy(input, target,  
weight=None, size_average=None, reduce=None,  
reduction='mean')[source]#
```

Parameters

Return type

Returns:

Tensor

Code Examples

```
>>> input = torch.randn(3, 2, requires_grad=True)
>>> target = torch.rand(3, 2, requires_grad=False)
>>> loss = F.binary_cross_entropy(torch.sigmoid(input), target)
>>> loss.backward()
```

Detailed Documentation

Documentation

Compute Binary Cross Entropy between the target and input probabilities.

See BCELoss for details.

input (Tensor) – Tensor of arbitrary shape as probabilities.

target (Tensor) – Tensor of the same shape as input with values between 0 and 1.

weight (Tensor, optional) – a manual rescaling weight if provided it's repeated to match input tensor shape

size_average (bool, optional) – Deprecated (see reduction).

reduce (bool, optional) – Deprecated (see reduction).

reduction (str, optional) – Specifies the reduction to apply to the output: 'none' | 'mean' | 'sum'. 'none': no reduction will be applied, 'mean': the sum of the output will be divided by the number of elements in the output, 'sum': the output will be summed. Note: size_average and reduce are in the process of being deprecated, and in the meantime, specifying either of those two args will override reduction. Default: 'mean'